RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS

Compute versus accuracy tradeoff in LSTM and transformers for medical video classification

Course Instructor:
Prof. Deepak Krishna Murthy
Professor of Practice
School of Computer Science and Engineering
RV University, Bangalore

Course Instructor:
Prof. Abdul Muqtadeer Ahmed
Assistant Professor
School of Computer Science and Engineering
RV University, Bangalore

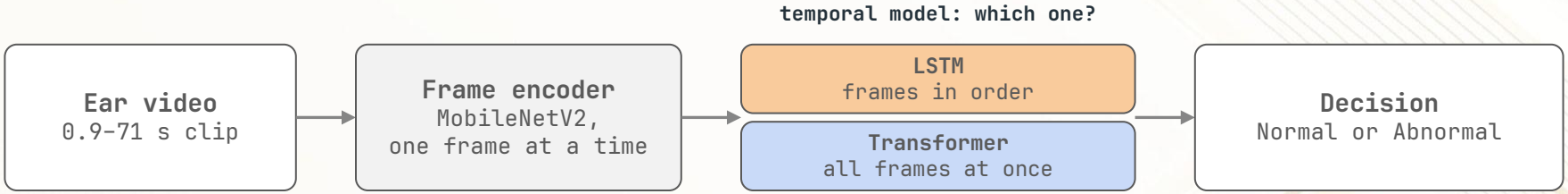
Team members
1RVU23CSE020 - Adhithi M
1RVU23CSE048 - Anagha M Mair
1RVU23CSE072 - Apoorv Gadiya
1RVU23CSE413 - Satyam Goyal

Problem Statement

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS

A handheld otoscope/camera records a few seconds of video of a child's ear canal.

Can an ordinary laptop CPU label each clip Normal or Abnormal, and which temporal model should it use?



Common belief
Transformers are more accurate but heavier. Does that hold for small medical video on a CPU?

H0-acc (accuracy)
Transformer AUROC = LSTM AUROC. We call them equal only if the gap stays within ± 0.03.

H0-cpu (compute)
For one clip at a time, the Transformer runs at least as fast as the LSTM on a CPU.

Approach: same input, head and splits; swap only the temporal model, then measure accuracy and CPU time.

Dataset: Pediatric Ear Videos

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS

874	26.4%	13 s	3–18
ear videos, all used	Abnormal (231 of 874)	median length (0.9–71 s)	age range in years

Class balance



One outer fold (x 6: 3 folds x 2 seeds)



77 Abnormal clips in every test fold; val only picks the stopping epoch and cut-offs.

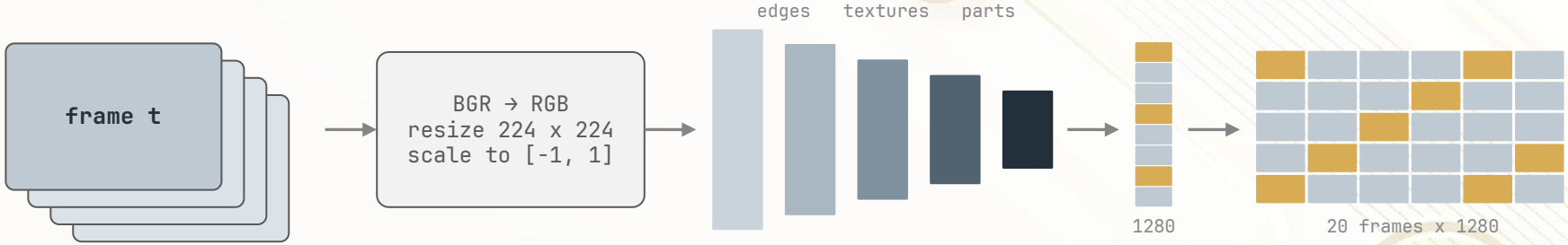
What we know about the data

- 480 x 640 portrait, 12.6–29.8 frames per second
- Labels file: video id, Normal or Abnormal, age
- Always saying Normal = 73.6% accuracy, so accuracy is never used to decide
- No patient ID: retakes of one child may sit in different folds
- Oral videos tried first: AUROC near 0.5, dropped

Takeaway: small and imbalanced: repeated cross-validation, and metrics that ignore the majority class.

Computer Vision: Frames to Features

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS



1. Sample
decode in order, keep 20 equally spaced frames

2. Prepare
what MobileNetV2 expects (old code used /255 and BGR)

3. MobileNetV2
frozen ImageNet CNN, run once per frame

4. Clip matrix
one 1280-number vector per frame, stacked

Why frozen
Run once for all 874 videos (5.5 min) and cached: every model gets identical input.

Why MobileNetV2
Built for phones: light depthwise convolutions, so it suits a CPU-only setup.

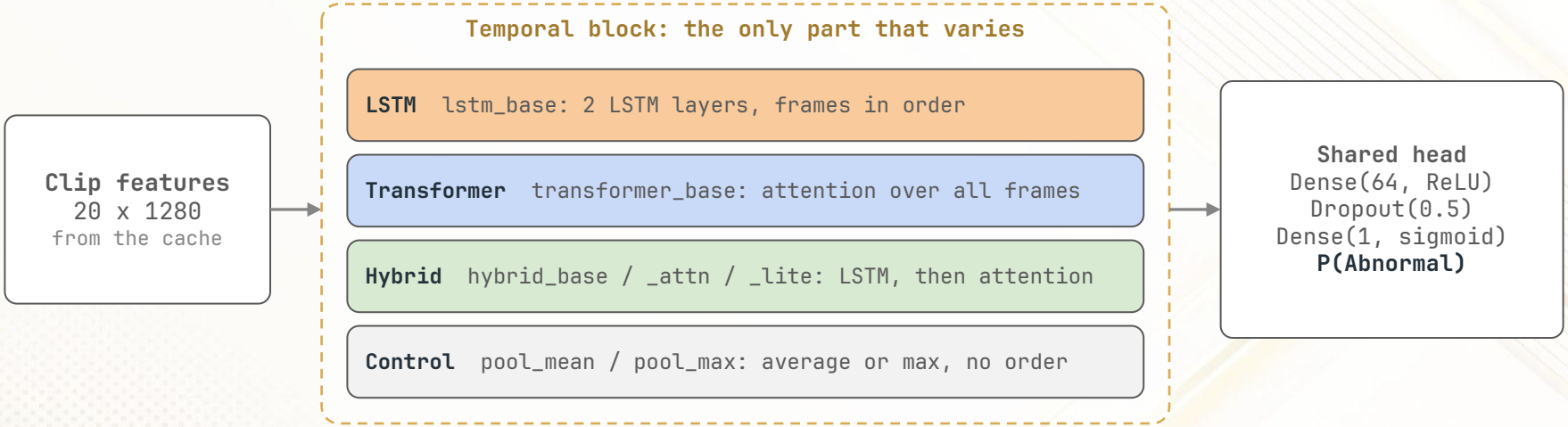
What it costs
312 ms per clip on 1 thread: the largest part of the pipeline (slide 10).

Training
Models learn only on the cached 20 x 1280 input: about 100x cheaper per run.

Takeaway: after this step a clip is a 20 x 1280 matrix; the temporal models never see pixels.

System Architecture

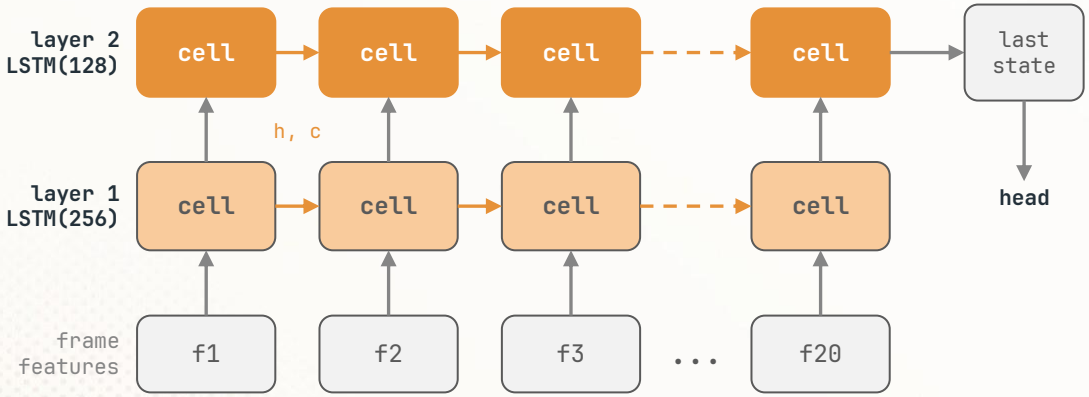
RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS



Held constant: features, splits, head, loss (cross-entropy with class weights), batch 32, stopping on validation AUROC (patience 15), seeds. **Varied:** the temporal block only.

Takeaway: any difference in accuracy or speed can only come from the temporal block.

LSTM: Reading Frames in Order



Step t needs h and c from step t-1: 20 steps run one after another.

Inside one cell

forget drop old memory	**input** write new info	**output** pass on

The cell state c carries memory across frames; the gates are small learned filters.

lstm_base

LSTM(256, all steps) → LSTM(128)
1.78 M parameters
35.4 M multiply-adds per clip
Adam, lr 1e-3, clipnorm 1
cost grows as T x d x d, in sequence

Takeaway: a natural fit for short clips, but the loop over frames cannot run in parallel.

Transformers: Self-Attention

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS

frame features (20, 1280)

↓

Dense 1280 → 256 + learned position

↓

2 x encoder block (pre-LN)

x + MultiHeadAttention(LN(x))

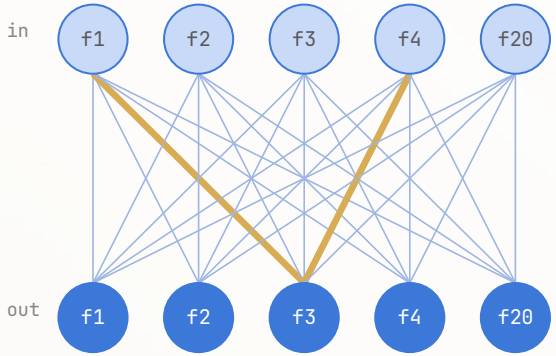
x + FeedForward(LN(x)), GELU

↓

LayerNorm → mean over 20 frames

↓

shared head → P(Abnormal)



Every frame scores every other frame: 20 x 20 = 400 pairs at once. Gold: output 3 draws mostly on frames 1 and 4.

$$\text{softmax}(Q \cdot K^T / \sqrt{d}) \cdot V$$

transformer_base

d 256, 4 heads, FFN 512
2 encoder blocks
1.40 M parameters (21% fewer)
28.0 M multiply-adds per clip
AdamW 3e-4, 5-epoch warm-up

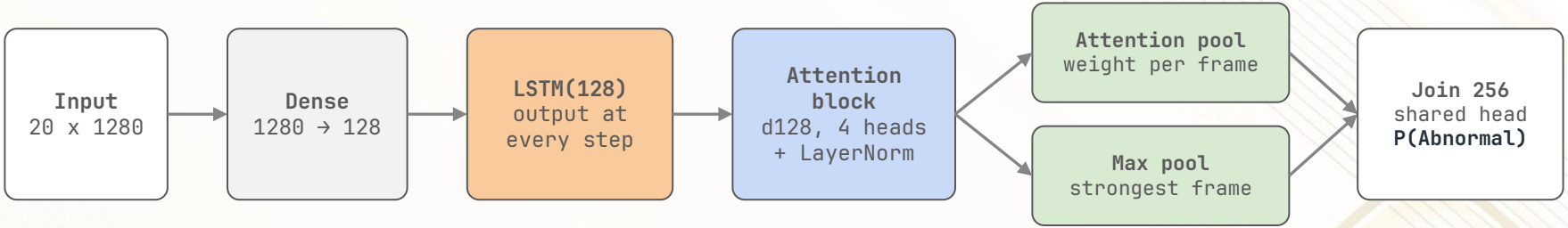
Why it suits a CPU

All 20 frames go through a few big matrix multiplications instead of 20 small dependent steps.
At 20 frames the pairwise (T x T) part is only about 3% of the work.

Takeaway: the Transformer sees all frames at once, so it needs no step-by-step loop.

Hybrid: How hybrid_attn Works

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS



Attention pooling, in one line

score_t = w · x_t + b, a = softmax over 20 frames,
summary = sum of a_t · x_t
A frame where the canal is clearly visible can get most of the weight.

	params	MACs	AUROC	ms
hybrid_base	0.86 M	17.2 M	0.905	2.12
hybrid_attn	**0.44 M**	**8.6 M**	**0.909**	**1.11**
hybrid_lite	0.16 M	3.0 M	0.900	0.92
lstm_base	1.78 M	35.4 M	0.907	3.54

Takeaway: hybrid_attn: highest AUROC (0.909), 1/4 of the LSTM's parameters, 3.2x its speed.

How We Tested

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS

1 **Extract**
20 frames → MobileNetV2, once, cached

2 **Split**
stratified 3-fold x 2 seeds, 15% val inside train

3 **Train**
7 models x 6 folds = 42 runs, about 18 min

4 **Time**
48 CPU jobs: 1 / 32 clips, 1 / 8 threads, 3 sessions

5 **Test**
corrected t-test and equivalence (±0.03)

What we score
AUROC (main): chance a random Abnormal clip ranks above a random Normal one; 0.5 = guessing.
Sensitivity at 90% specificity: Abnormal clips caught when 1 in 10 Normal clips is flagged.
Accuracy is reported, never used to decide.

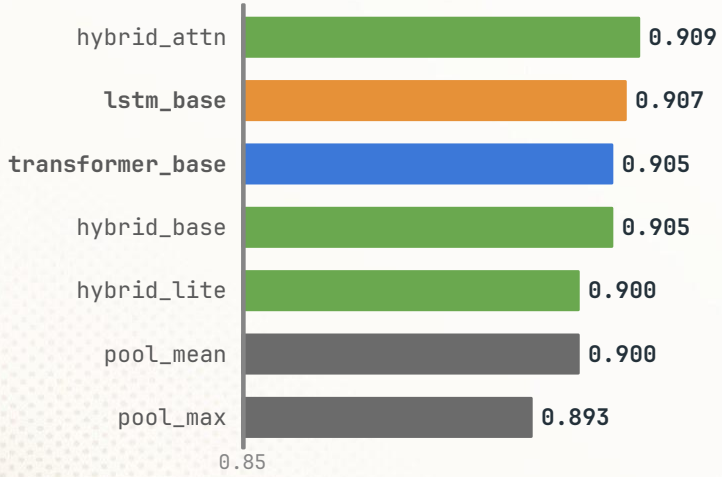
How we time
Compiled tf.function, 50 warm-up + 300 timed calls, median of 3 sessions, new process each time.
Intel i5-12450H laptop, 15.7 GB RAM, no GPU; temporal model and full pipeline timed.

Takeaway: every model is scored on the same 6 test folds, so they are compared fold by fold.

Results: Accuracy

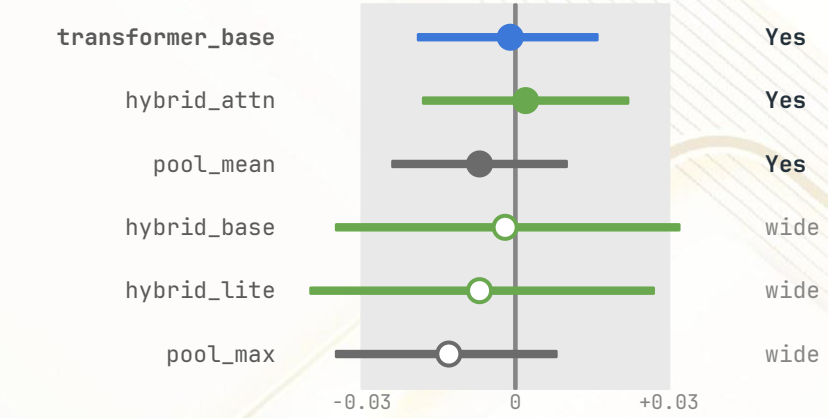
RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS

Test AUROC, mean of 6 folds
axis starts at 0.85 to show the small gaps



for scale: guessing gives AUROC 0.5

Difference from lstm_base (90% CI)
grey band = equivalence margin ± 0.03



Corrected t-test: no model differs from lstm_base (all p ≥ 0.26).
Filled = equivalent.

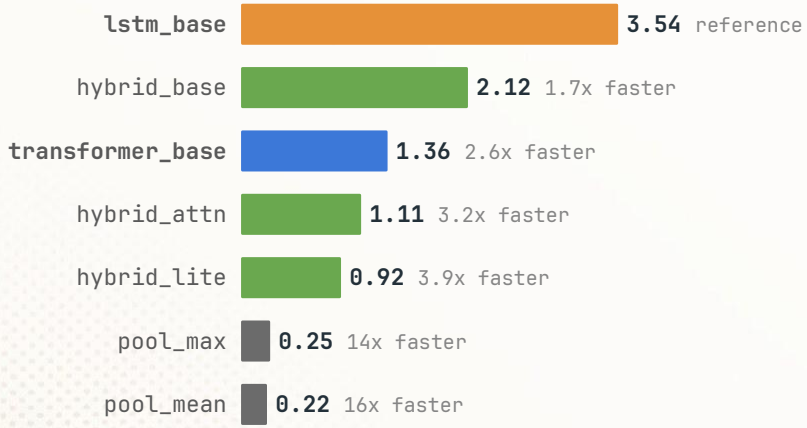
H0-acc supported: Transformer - LSTM = -0.001 AUROC, inside ±0.03; all 7 models within fold noise.

Results: CPU Cost

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS

Temporal model + head, ms per clip

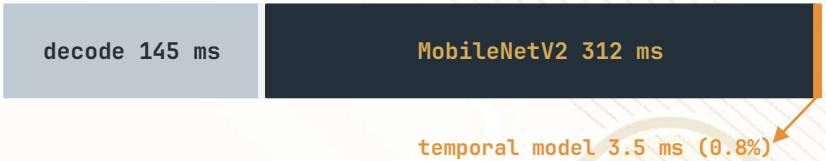
one clip at a time, 1 CPU thread, median of 3 sessions



32 clips on 8 threads: LSTM 18.9 ms, Transformer 19.1 ms (a tie).
Peak RAM 330–410 MB for every model.

Where the time goes, one clip end to end

1 CPU thread, about 465 ms in total with lstm_base



2.6x
Transformer faster than LSTM, 1 clip, 3 of 3 sessions

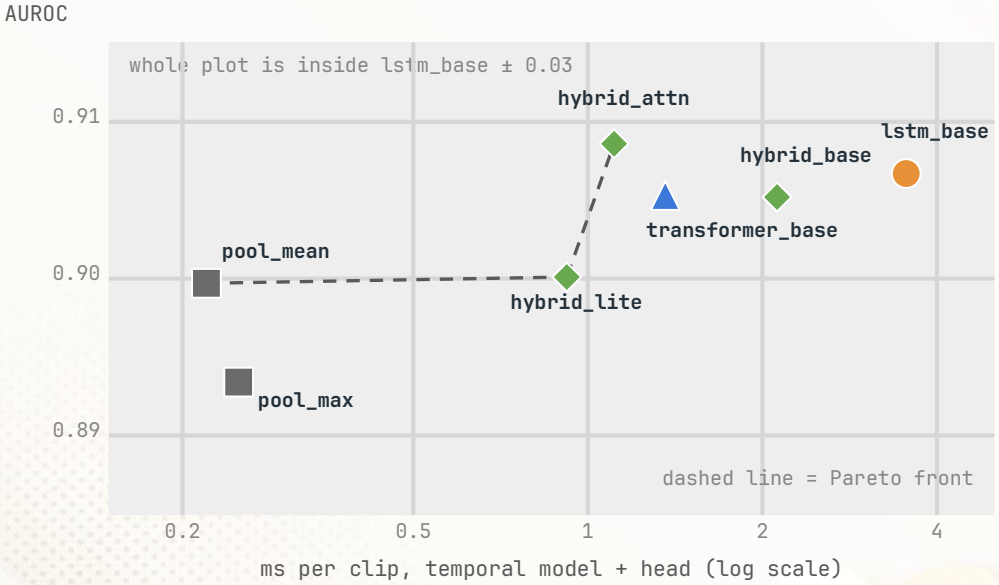
Tie
LSTM and Transformer at 32 clips on 8 threads

<1%
of the clip time is the temporal model

H0-cpu supported: the Transformer is cheaper, yet decoding + MobileNetV2 are over 99% of the time.

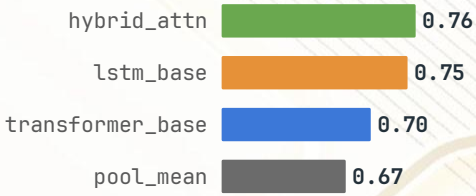
Results: Accuracy vs Cost

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS



Screening view

sensitivity at 90% specificity



Cut-off set on validation for 95% recall: test recall 0.87–0.92, 6–10 of 77 Abnormal missed per fold.

Best overall
hybrid_attn
0.909 AUROC, 1.11 ms

Cheapest
pool_mean
0.900 AUROC, 0.22 ms

Takeaway: accuracy is tied, so cost decides: hybrid_attn for screening, pool_mean for lowest cost.

Conclusions and Next Steps

RV UNIVERSITY
Go, change the world
an initiative of RV EDUCATIONAL INSTITUTIONS

H0-acc: Supported
LSTM 0.907 vs Transformer 0.905; 90% CI inside ± 0.03

H0-cpu: Supported
Transformer 1.36 ms vs LSTM 3.54 ms (2.6x faster)

What we found

- No trade-off: the Transformer is as accurate and cheaper
- Frame order adds little: a plain average reaches 0.900
- The temporal model is < 1% of the time per clip
- hybrid_attn: highest AUROC (0.909) and sensitivity (0.76)

What to use

- Transformer over LSTM on a CPU
- hybrid_attn for best sensitivity; pool_mean for lowest cost
- Pick the cut-off on validation data, not 0.5
- Score single clips on 1 thread (8 threads: 1.4–1.8x slower)

Limits and next

- 6 folds, 20 frames, one dataset, one laptop, no patient ID
- Not a validated diagnostic tool
- Next: fine-tuned or medical frame encoder, 10 / 40 frames, external test set

Bottom line: the frame encoder, not the temporal model, is the bottleneck; improve the features next.